

MBAi 448 | Winter 2026

Aligned AI

Today

- Assignment 6 / week 7 takeaways *10 minutes*
- Technical alignment *15 minutes*
- Evaluating alignment *15 minutes*
- Quick break *5 minutes*
- Generative AI case studies *20 minutes*
- Assignment walkthrough *20 minutes*

Assignment 6 / week 7 takeaways

Aligned AI

Week 7 takeaways (via ChatGPT)



➤ **Model Performance $\neq$ Business Value**

Traditional metrics (accuracy, precision/recall, MSE) are insufficient for **generative AI** evaluating the impact of systems in real world. Modern systems must also optimize for:

- Instruction-following
- Groundedness and citation correctness
- Safety and fairness
- Latency and cost per successful task

MBA Insight: Executives must define success metrics aligned to business outcomes (trust, compliance, efficiency), not just model accuracy.

Week 7 takeaways (via ChatGPT)

➤ Alignment Is a Strategic Control Mechanism

Techniques like supervised fine-tuning, instruction tuning, RLHF, and RAG are not just technical upgrades — they are governance tools that shape model behavior.

MBA Insight: Alignment is about risk management, brand protection, and user trust. Poor alignment - or poor usage - can create legal, reputational, and operational exposure.

➤ Optimization Can Backfire (Goodhart's Law in AI)

The chatbot activity illustrates that optimizing too aggressively for a scorecard can lead to gaming the metric rather than serving the customer.

MBA Insight: When incentives are misaligned, AI systems can optimize for measurable proxies rather than real customer value — a familiar challenge in performance management.

?

Week 7 takeaways (via ChatGPT)

➤ **RAG Systems Fail in Predictable Ways**

- Weak retrieval (missing or outdated index), poor chunking, or too much context
- Hallucinations or alignment gaps
- Lack of attribution and observability

MBA Insight: AI failures are ~~often~~ **always** architectural and operational — not ~~just~~ “model intelligence” problems. Leaders must invest in infrastructure, data quality, and monitoring.

➤ **Benchmarks and Leaderboards Don't Equal Production Readiness**

Benchmarks like Hugging Face leaderboards or academic tests (e.g., MMLU, HumanEval) signal capability, but they don't capture enterprise constraints like compliance, latency, or integration complexity.

MBA Insight: Competitive advantage comes from deployment quality, not benchmark rankings.

Week 7 takeaways (via ChatGPT)

➤ **AI Systems Are Socio-Technical Systems**

Production AI requires:

- Evaluation frameworks
- Observability and monitoring
- UX/UI and workflow integration
- Governance and risk controls

MBA Insight: ~~AI transformation~~ **Technology implementation** is organizational, not just technical. Success requires cross-functional coordination across product, engineering, legal, compliance, and operations.

Technical alignment

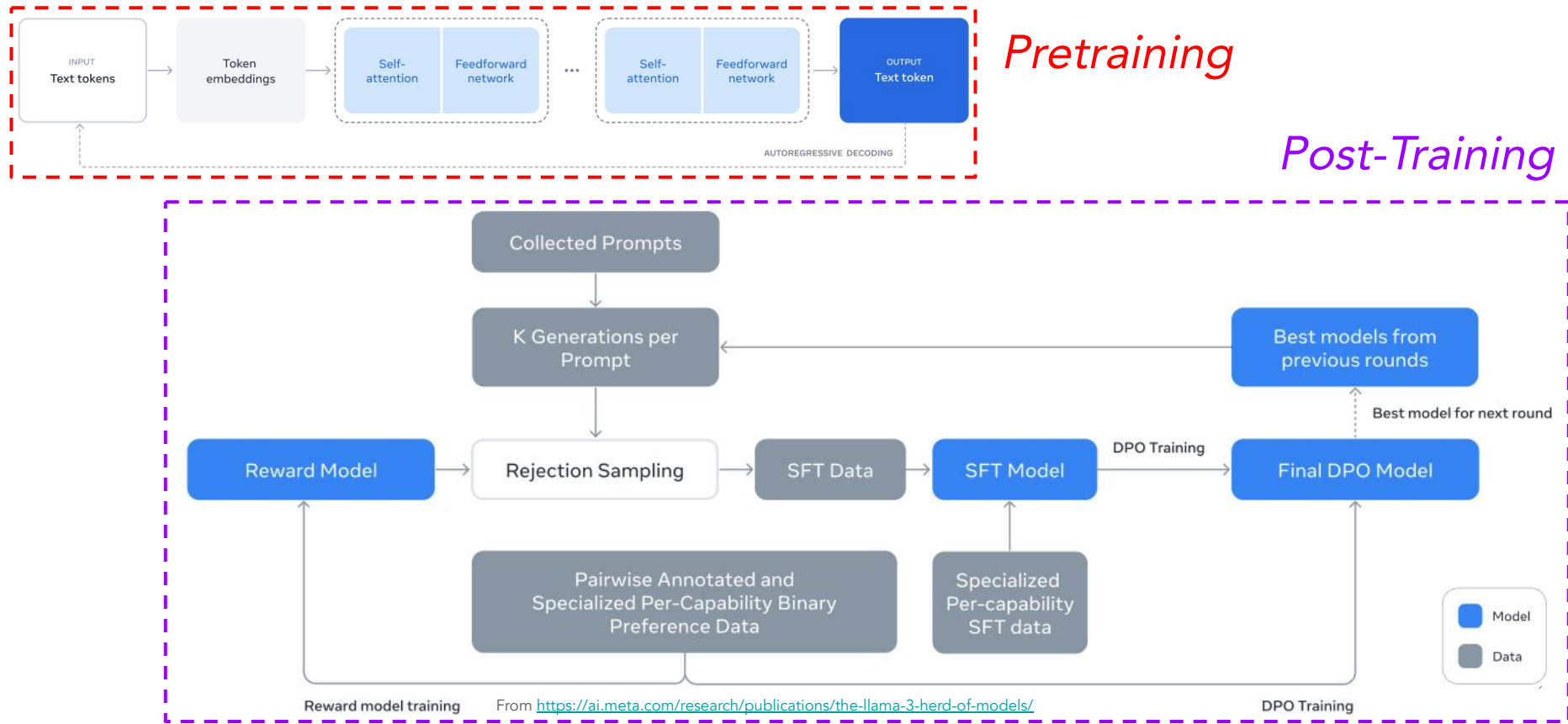
Aligned AI

There are different phases in LLM training

Model tree for allenai/Llama-3.1-Tulu-3-70B ⓘ



There are different phases in LLM training



Alignment starts before the model

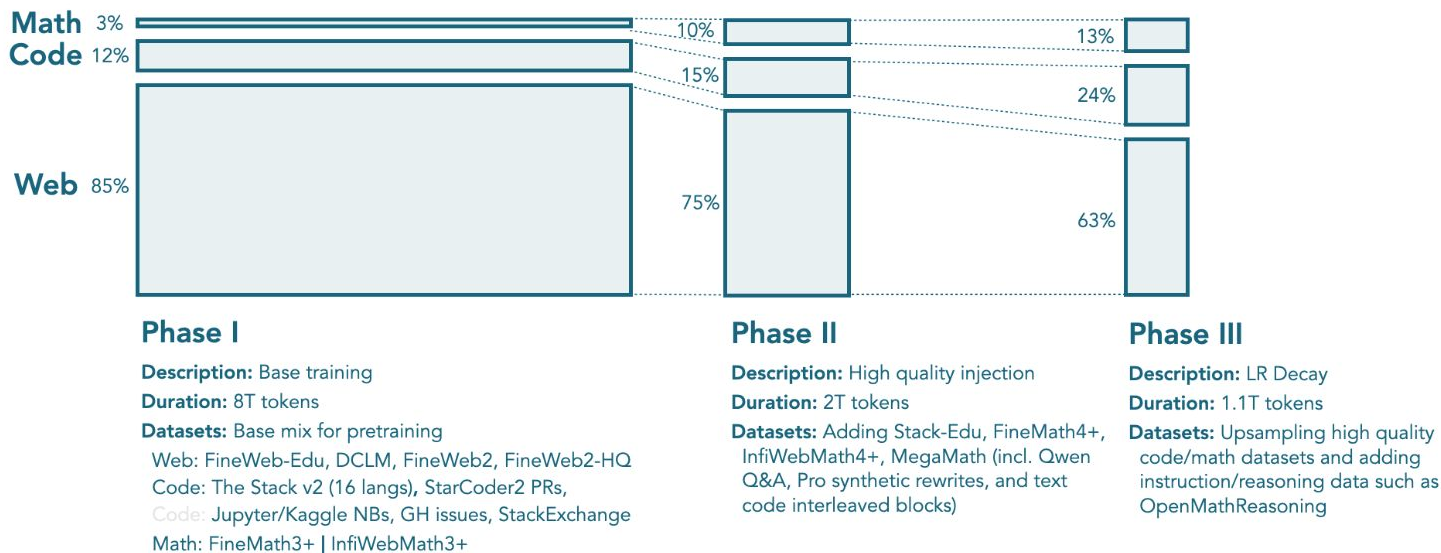


X to stop Grok AI from undressing images of real people after backlash

"With NSFW (not safe for work) settings enabled, Grok is supposed to allow "upper body nudity of imaginary adult humans (not real ones)" consistent with what can be seen in R-rated films, Musk wrote online on Wednesday."

Pretraining pipelines are carefully structured

Pretraining Recipe



LR Schedule



Post-training pipelines are carefully structured

Reinforcement learning objectives. We use a variety of reward functions to improve helpfulness, math, coding, reasoning, instruction-following, and multilingual abilities, while minimizing model harmfulness. This includes learning from weight averaged reward models (Ramé et al., 2024b) trained with human feedback data, code execution feedback (Gehring et al., 2024), and ground-truth rewards for solving math problems (DeepSeek-AI, 2025; Lambert et al., 2024).

Data filtering. We carefully optimize the data used in post-training to maximize model performance. We filter examples that show certain personal information, unsafe or toxic model outputs, mistaken self-identification data, and duplicated examples. Including subsets of data that encourage better in-context attribution, hedging, and refusals to minimize hallucinations also improves performance on factuality metrics, without degrading model performance on other metrics.

Category	Prompt Dataset	Count	# Prompts used in SFT	# Prompts used in DPO	Reference
General	Tulu 3 Hardcoded [†]	24	240	—	—
	OpenAssistant ^{1,2,4}	88,838	7,132	7,132	Köpf et al. (2024)
	No Robots	9,500	9,500	9,500	Rajani et al. (2023)
	WildChat (GPT-4 subset) ⁴	241,307	100,000	100,000	Zhao et al. (2024)
	UltraFeedback ^{α,2}	41,635	—	41,635	Cui et al. (2023)
Knowledge	FLAN v2 ^{1,2,4}	89,982	89,982	12,141	Longpre et al. (2023)
Recall	SciRIFF ⁴	35,357	10,000	17,590	Wadden et al. (2024)
	TableGPT ⁴	13,222	5,000	6,049	Zha et al. (2023)
Math	Tulu 3 Persona MATH	149,960	149,960	—	—
Reasoning	Tulu 3 Persona GSM	49,980	49,980	—	—
	Tulu 3 Persona Algebra	20,000	20,000	—	—
	OpenMathInstruct 2 ⁴	21,972,791	50,000	26,356	Toshniwal et al. (2024)
	NuminaMath-TIR ^α	64,312	64,312	8,677	Beeching et al. (2024)
Coding	Tulu 3 Persona Python	34,999	34,999	—	—
	Evol CodeAlpaca ^α	107,276	107,276	14,200	Luo et al. (2023)
Safety & Non-Compliance	Tulu 3 CoCoNot	10,983	10,983	10,983	Brahman et al. (2024)
	Tulu 3 WildJailbreak ^{α,4}	50,000	50,000	26,356	Jiang et al. (2024)
	Tulu 3 WildGuardMix ^{α,4}	50,000	50,000	26,356	Han et al. (2024)
Multilingual	Aya ⁴	202,285	100,000	32,210	Singh et al. (2024b)
Precise IF	Tulu 3 Persona IF	29,980	29,980	19,890	—
	Tulu 3 IF-augmented	65,530	—	65,530	—
Total		23,327,961	939,344	425,145 ^γ	

Post-training shapes how the model acts

2.2 Data

For code agents, we want to foster an interaction pattern in which the agent, and consequently the underlying model, alternates between chain-of-thought [Wei et al., 2022] reasoning and actions within the coding environment to perform the software engineering task. We generate supervised trajectories by running an agent in the SWE-Gym environments [Pan et al., 2024] with the OpenHands CodeAct scaffold [Wang et al., 2025, 2024]. We execute unit tests to determine the final patch quality.

We also include a selected mixture of natural language data to retain general natural language understanding capabilities in Devstral-Small.

2.3 Post-training

We employ a two-stage training process. Initially, we train the model on a larger subset of rollouts that meet our first level of quality standards, determined by a heuristic filter. In the second stage, we fine-tune the model using only the trajectories that pass our strictest filters.

We then perform additional rounds of rollouts with the finetuned model, and train further the model with policy optimization.

Models learn a policy to steer responses

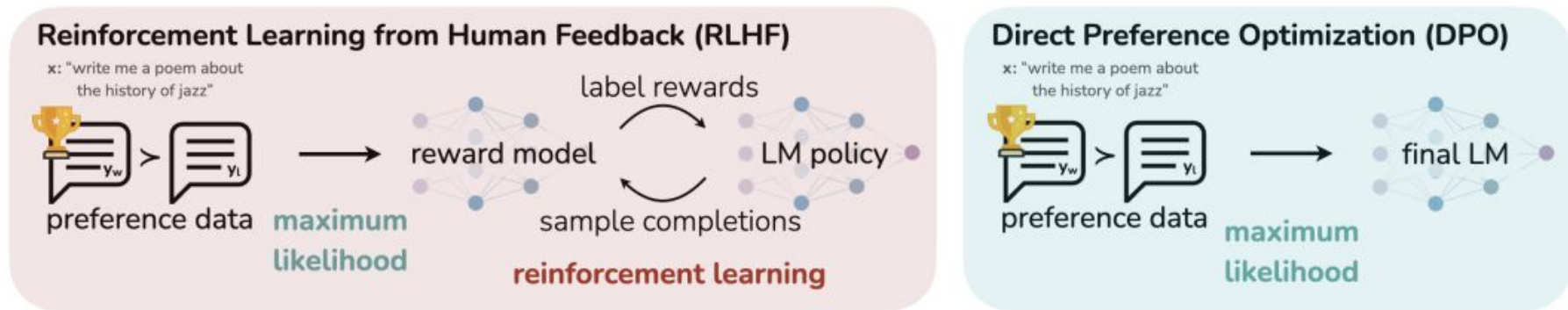


Figure 1: **DPO optimizes for human preferences while avoiding reinforcement learning.** Existing methods for fine-tuning language models with human feedback first fit a reward model to a dataset of prompts and human preferences over pairs of responses, and then use RL to find a policy that maximizes the learned reward. In contrast, DPO directly optimizes for the policy best satisfying the preferences with a simple classification objective, fitting an *implicit* reward model whose corresponding optimal policy can be extracted in closed form.

Models learn a policy to steer responses

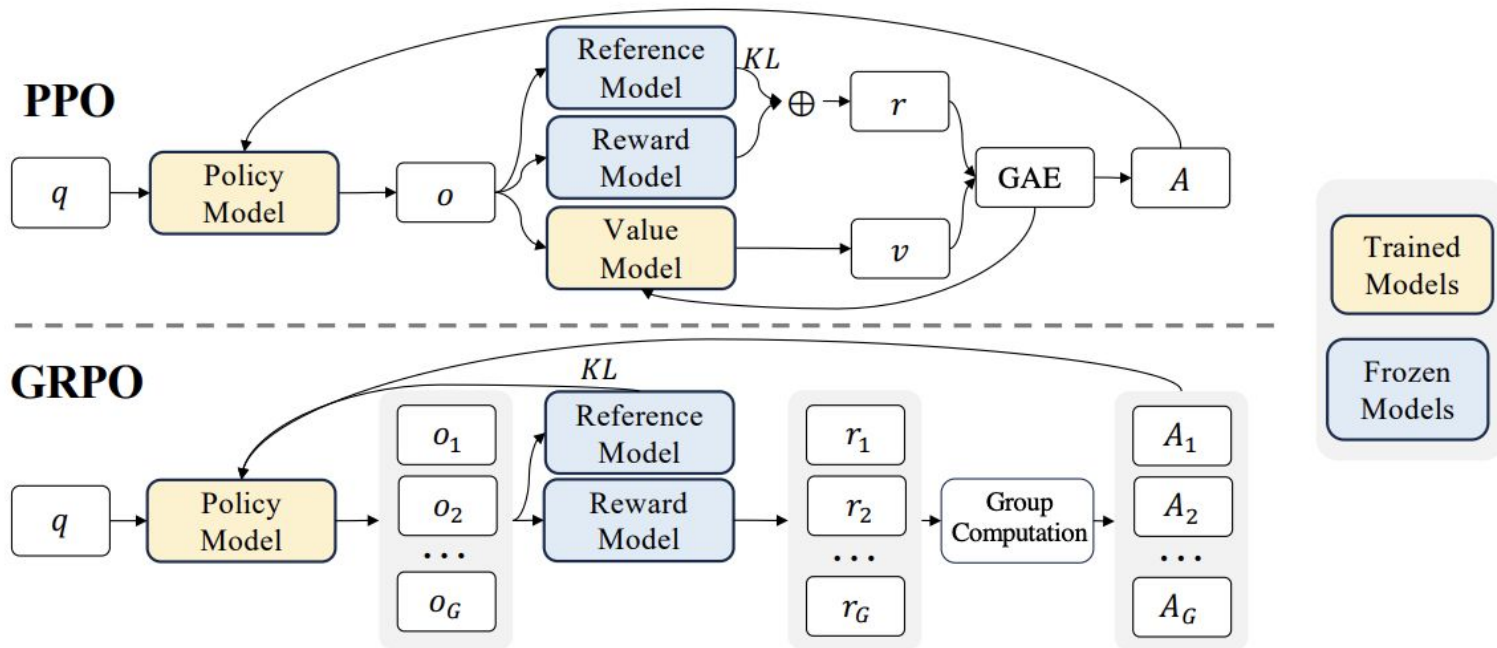
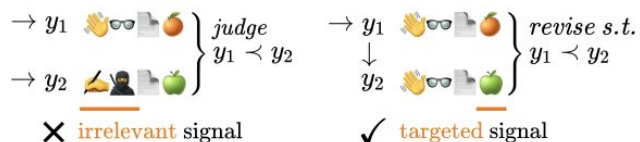


Figure 4 | Demonstration of PPO and our GRPO. GRPO foregoes the value model, instead estimating the baseline from group scores, significantly reducing training resources.

Models learn a policy to steer responses

(A) x write a story about apples

Judged preferences (RLAIF): Contrastive revisions (CLAIR):



(B) Different alignment situations for model π_θ

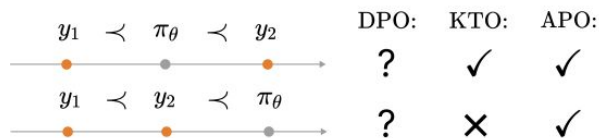


Figure 1: Alignment is underspecified with regard to preferences and training objective. **A:** Preference pairs can vary along irrelevant aspects, Contrastive Learning from AI Revisions (CLAIR) creates a targeted preference signal instead. **B:** The quality of the model can impact alignment training, Anchored Preference Optimization (APO) explicitly accounts for this.

$$(i) \nabla_\theta \mathcal{L}_{DPO}(x, y_w, y_l; \theta) =$$

$$-\beta \left[\nabla_\theta \left(\log \pi_\theta(y_w | x) - \log \pi_\theta(y_l | x) \right) \sigma \left(r_\theta(x, y_l) - r_\theta(x, y_w) \right) \right]$$

increase **likelihood difference** more **when losing reward is greater**

$$(ii) \nabla_\theta \mathcal{L}_{zero}^{APO}(x, y_w, y_l; \theta) =$$

$$-\beta \left[\nabla_\theta \left(\log \pi_\theta(y_w | x) \right) \delta \left(r_\theta(x, y_w) \right) - \nabla_\theta \left(\log \pi_\theta(y_l | x) \right) \delta \left(r_\theta(x, y_l) \right) \right]$$

increase **winning likelihood** decrease **losing likelihood**
more when **winning reward close to 0** more when **losing reward close to 0**

$$(iii) \nabla_\theta \mathcal{L}_{down}^{APO}(x, y_w, y_l; \theta) =$$

$$-\beta \left[-\nabla_\theta \left(\log \pi_\theta(y_w | x) \right) \delta \left(r_\theta(x, y_w) \right) + \nabla_\theta \left(\log \pi_\theta(y_w | x) - \log \pi_\theta(y_l | x) \right) \delta \left(r_\theta(x, y_w) - r_\theta(x, y_l) \right) \right]$$

decrease **winning likelihood** increase **likelihood difference**
more when **winning reward close to 0** more when **reward difference close to 0**

Models learn a policy to steer responses

Post-training Recipe

The post-training recipe starts of with the checkpoint after pretraining and long context adaptation. The optimizer state is not re-used from earlier training.

Pretrained + Long context

Model Merging

Finally, we linearly merge the model soup checkpoint with long context checkpoint using a 90/10 ratio to recover the long context capabilities.

During mid-training the model is trained for **5 epochs** on a mix of OpenThoughts and Nemotron post-training data totalling **175B tokens** (35B unique).

Mid-training

Model Soup

We found that model souping, where intermediate checkpoints from the APO stage are merged together further improves the downstream performance.

The model is further trained on a mix of **25 high quality datasets** mixing reasoning/non-reasoning data for **4 epochs** and **10B tokens** (2.5B unique).

SFT

APO

We apply Anchored Preference Optimization (APO), a variant of DPO, for 1 epoch with a thinking preference pair for every non-thinking preference pair.

Short SFT
10%
90%

→

↓

←←←←←

Technical alignment is not all you need

How we're addressing sycophancy

Beyond rolling back the latest GPT-4o update, we're taking more steps to realign the model's behavior:

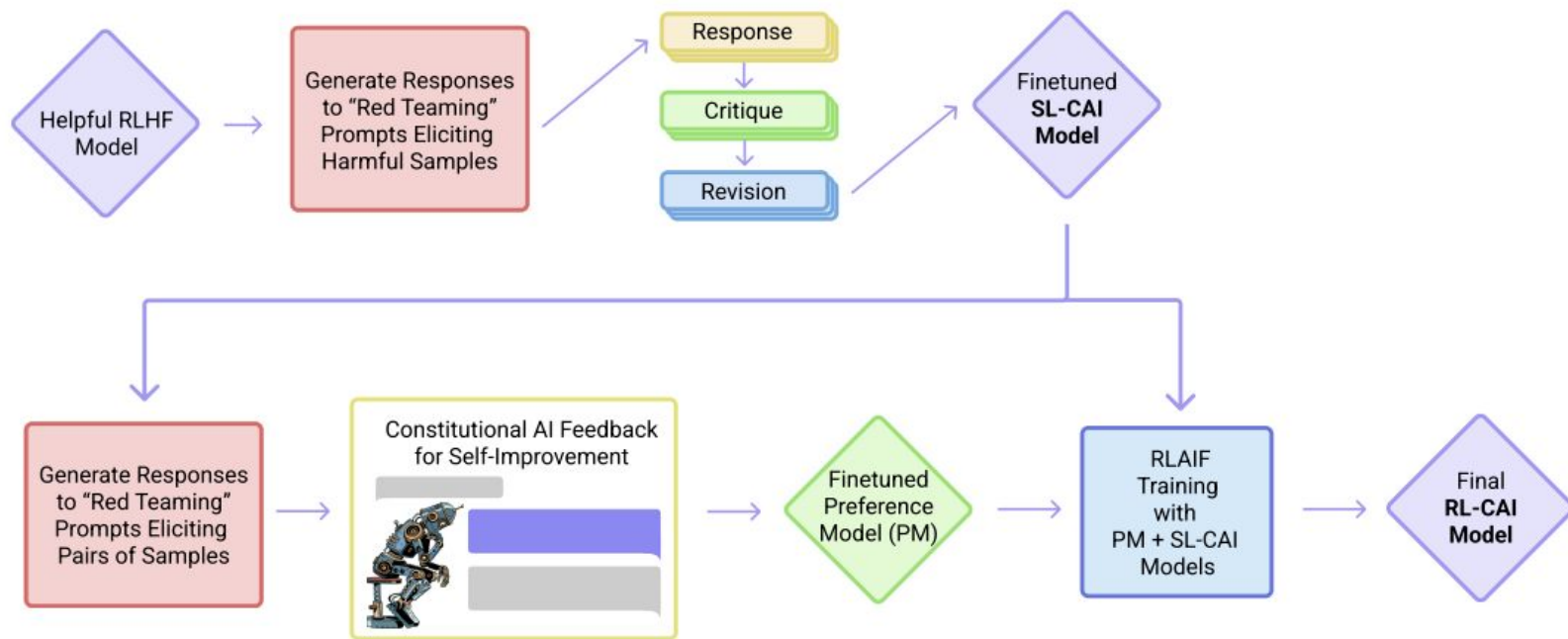
- Refining core training techniques and system prompts to explicitly steer the model away from sycophancy.
- Building more guardrails to increase honesty and transparency—principles in our Model Spec.
- Expanding ways for more users to test and give direct feedback before deployment.
- Continue expanding our evaluations, building on the Model Spec and our ongoing research, to help identify issues beyond sycophancy in the future.

We also believe users should have more control over how ChatGPT behaves and, to the extent that it is safe and feasible, make adjustments if they don't agree with the default behavior.

Evaluating alignment

Aligned AI

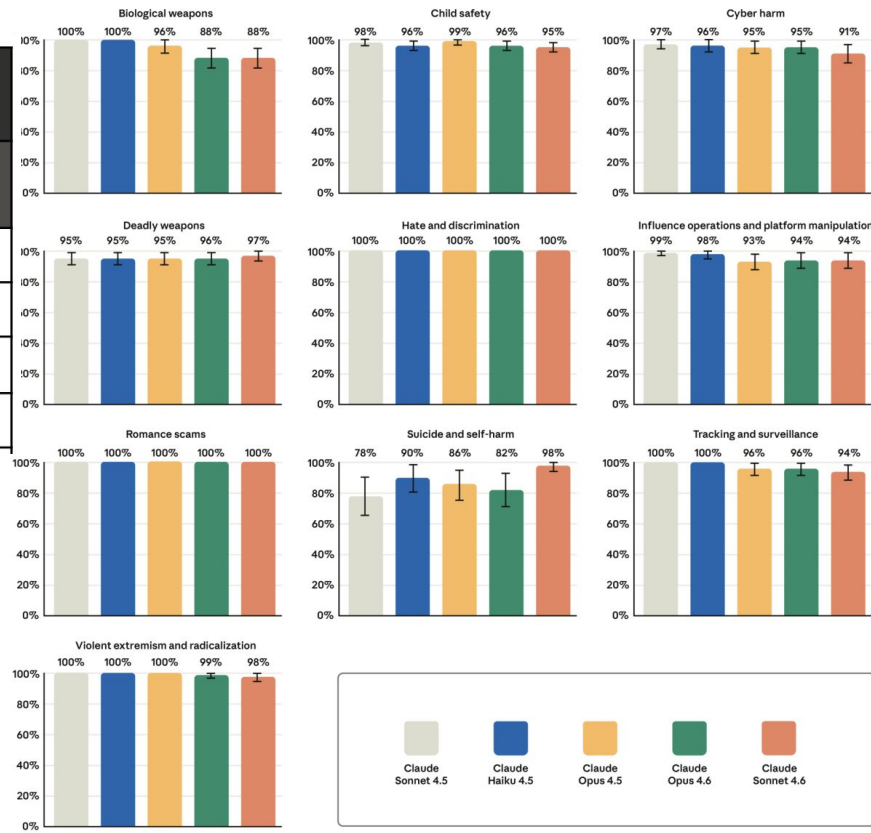
Technical alignment will only take you so far



Technical alignment will only take you so far

Multi-Turn Conversation Evaluations

Model		Attack success rate without safeguards		Attack success rate with safeguards	
		1 attempt	200 attempts	1 attempt	200 attempts
Claude Sonnet 4.6	Extended thinking	12.0%	42.9%	8.0%	50.0%
	Standard thinking	14.4%	64.3%	8.6%	50.0%
Claude Opus 4.6	Extended thinking	17.8%	78.6%	9.7%	57.1%
	Standard thinking	20.0%	85.7%	10.0%	64.3%



Alignment and effectiveness are intertwined

High level overview of AI Safety Levels (ASLs)

ASL-1

Smaller
models

ASL-2

Present
large models

ASL-3

Significantly
higher risk

ASL-4+

Speculative

Increasing model capability,
Increasing security and safety measures

Alignment and effectiveness are intertwined

- ASL-1 refers to systems which pose no meaningful catastrophic risk, for example a 2018 LLM or an AI system that only plays chess.
- ASL-2 refers to systems that show early signs of dangerous capabilities – for example ability to give instructions on how to build bioweapons – but where the information is not yet useful due to insufficient reliability or not providing information that e.g. a search engine couldn't. Current LLMs, including Claude, appear to be ASL-2.
- ASL-3 refers to systems that substantially increase the risk of catastrophic misuse compared to non-AI baselines (e.g. search engines or textbooks) OR that show low-level autonomous capabilities.
- ASL-4 and higher (ASL-5+) is not yet defined as it is too far from present systems, but will likely involve qualitative escalations in catastrophic misuse potential and autonomy.

GPT-2 was once considered dangerous

For that reason, OpenAI said, it's only releasing a smaller version of the language model, citing [its charter](#), which noted that the organization expects that "safety and security concerns will reduce our traditional publishing in the future." Admittedly, the organization said that it wasn't sure of the decision, "we believe that the AI community will eventually need to tackle the issue of publication norms in a thoughtful way in certain research areas."

Not everyone took that well. OpenAI's [tweet](#) announcing GPT-2 was met with anger and frustration, accusing the company of "closing off" its research, and doing the "opposite of open," seizing on the company's name.

Others [were more forgiving](#), calling the move a "new bar for ethics" for thinking ahead of possible abuses.

Jack Clark, policy director at OpenAI, said the organization's priority is "not enabling malicious or abusive uses of the technology," calling it a "very tough balancing act for us."

GPT-2 was once considered a frontier model



karpathy

3 weeks ago

Maintainer

edited



When OpenAI released GPT-2 in February 2019, training the largest model (1.5B parameters) required serious compute:

- **Hardware:** 32 TPU v3 chips (256 TPU v3 cores, 8 cores per chip)
- **Training time:** "A bit over a week" (~168 hours)
- **Cloud cost:** At \$8/hour per TPU v3, that's $32 \times 168 \times \$8 =$ **\$43,000**

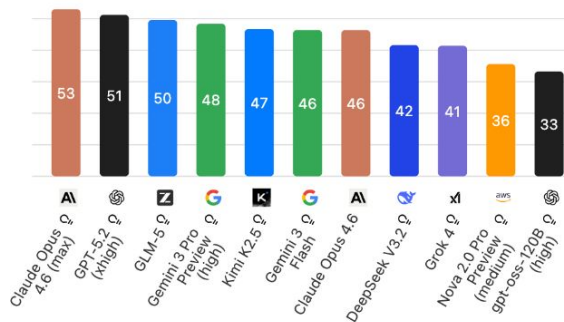
Sources: [Reddit thread from 2019](#), [HuggingFace model card](#).

Beating GPT-2 for <\$100 from scratch has been a bit of an odd obsession for me but finally here we are. Seven years later, we can beat GPT-2's performance in nanochat ~1000 lines of code running on a single 8XH100 GPU node for ~3 hours. At ~\$24/hour for an 8xH100 node, that's \$73, i.e. ~600x cost reduction. That is, each year the cost to train GPT-2 is falling to approximately 40% of the previous year. (I think this is an underestimate and that further improvements are still quite possible). The gains come from everywhere: better hardware (H100 vs TPU v3), better software (Flash Attention 3, torch.compile), better algorithms (Muon optimizer, architectural improvements), and better data (FineWeb-edu).

How can you tell how capable a model is?

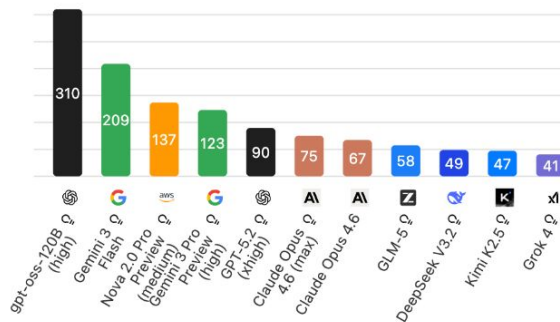
INTELLIGENCE

Artificial Analysis Intelligence Index; Higher is better



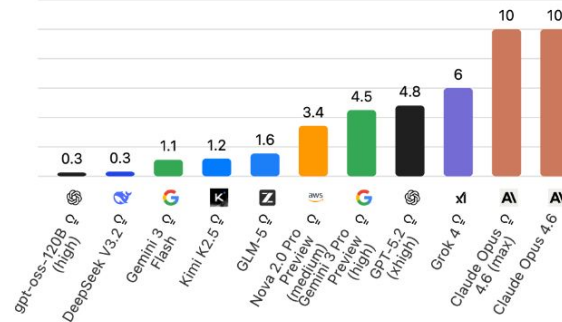
SPEED

Output Tokens per Second; Higher is better



PRICE

USD per 1M Tokens; Lower is better

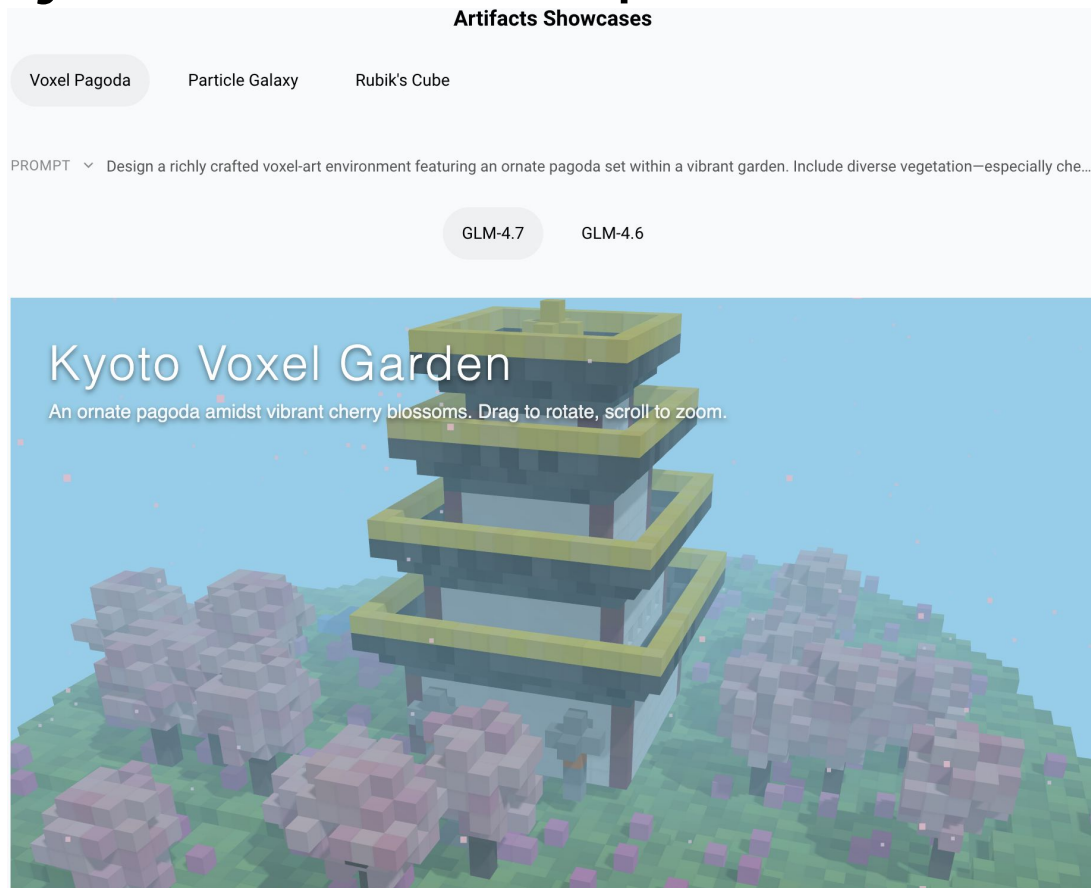


How can you tell how capable a model is?

	Category	Benchmark	CoT	# Shots	Chat	Multiturn ICL	Metric
<i>Development</i>	Knowledge Recall	MMLU	✓	0	✓	✗	EM
		PopQA	✗	15	✓	✓	EM
		TruthfulQA	✗	6	✓	✗	MC2
	Reasoning	BigBenchHard	✓	3	✓	✓	EM
		DROP	✗	3	✗	N/A	F1
	Math	GSM8K	✓	8	✓	✓	EM
		MATH	✓	4	✓	✓	Flex EM
	Coding	HumanEval	✗	0	✓	N/A	Pass@10
		HumanEval+	✗	0	✓	N/A	Pass@10
	Instruction Following	IFEval	✗	0	✓	N/A	Pass@1 (prompt; loose)
		AlpacaEval 2	✗	0	✓	N/A	LC Winrate
	Safety	TÜLU 3 Safety	✗	0	✓	N/A	Average*
<i>Unseen</i>	Knowledge Recall	MMLU-Pro	✓	0	✓	N/A	EM
		GPQA	✓	0	✓	N/A	EM
	Reasoning	AGIEval English	✓	0	✓	✓	EM
	Math	Deepmind Mathematics	✓	0	✓	✓	EM (Sympy)
	Coding	BigCodeBench	✗	0	✓	N/A	Pass@10
	Instruction Following	IFEval-OOD	✗	0	✓	N/A	Pass@1 (prompt; loose)
		HREF	✗	0	✓	N/A	Winrate

Table 24 The TÜLU 3 Evaluation Regime: settings for development (**top**) and unseen (**bottom**) portions of the evaluation suite. **CoT** are evaluations run with chain of thought prompting (Wei et al., 2022b). **#Shots** is the number of in-context examples in the evaluation template. **Chat** refers to whether we use a chat template while prompting the model. **Multiturn ICL** refers to a setting where we present each in-context example as a separate turn in a conversation (applicable only when a chat template is used and # Shots is not 0). *Average over multiple sub-evaluations – full details of the safety evaluation are included in the Appendix.

How can you tell how capable a model is?



How can you tell how capable a model is?



ARC-AGI-2 LEADERBOARD

AGI REMAINS UNSOLVED.
NEW IDEAS STILL NEEDED.

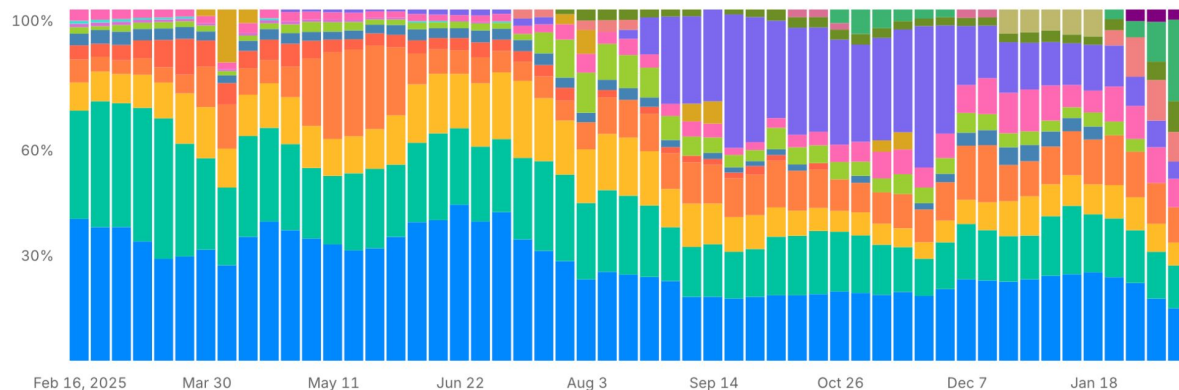
The chart displays the performance of various AI models on the ARC-AGI-2 benchmark. The Y-axis represents the Score (%) from 0% to 100%. The X-axis represents the Cost per Task (\$) on a logarithmic scale from \$1e-3 to \$100. A legend indicates that models with a colored square are 'ARC PRIZE' and models with a triangle are 'VERIFIED'. The chart shows a general trend of increasing score with increasing cost, with some models achieving higher scores at lower costs than others. The date 02/19/2026 is noted in the bottom right corner.

Model	Score (%)	Cost per Task (\$)	ARC PRIZE	VERIFIED
Gemini 3 Deep Think (2/26)	~85	~\$10	Yes	Yes
GPT-5.2 (Refine.)	~75	~\$10	No	Yes
Claude Opus 4.6 (120K, Medium)	~65	~\$1	Yes	No
Claude Sonnet 4.6 (High)	~60	~\$1	Yes	No
GPT-5.2 Pro (High)	~55	~\$10	No	Yes
GPT-5.2 (X-High)	~50	~\$1	No	Yes
GPT-5.2 (High)	~45	~\$1	No	Yes
GPT-5.2 Pro (Medium)	~40	~\$10	No	Yes
Gemini 3 Flash Preview (High)	~35	~\$0.1	Yes	No
NVARC	~30	~\$0.1	Yes	No
GPT-5.2 (Medium)	~25	~\$1	No	Yes
Opus 4.5 (Thinking, 16K)	~20	~\$1	Yes	No
GPT-5 Pro	~15	~\$10	No	Yes
Claude Sonnet 4.5 (Thinking 32K)	~10	~\$1	Yes	No
GPT-5 (High)	~5	~\$1	No	Yes
Claude Opus 4 (Thinking 16K)	~5	~\$1	Yes	No
GPT-4.5	~5	~\$1	No	Yes
G3-Pro (Medium)	~5	~\$10	No	Yes

How can you tell how capable a model is?

📊 Market Share

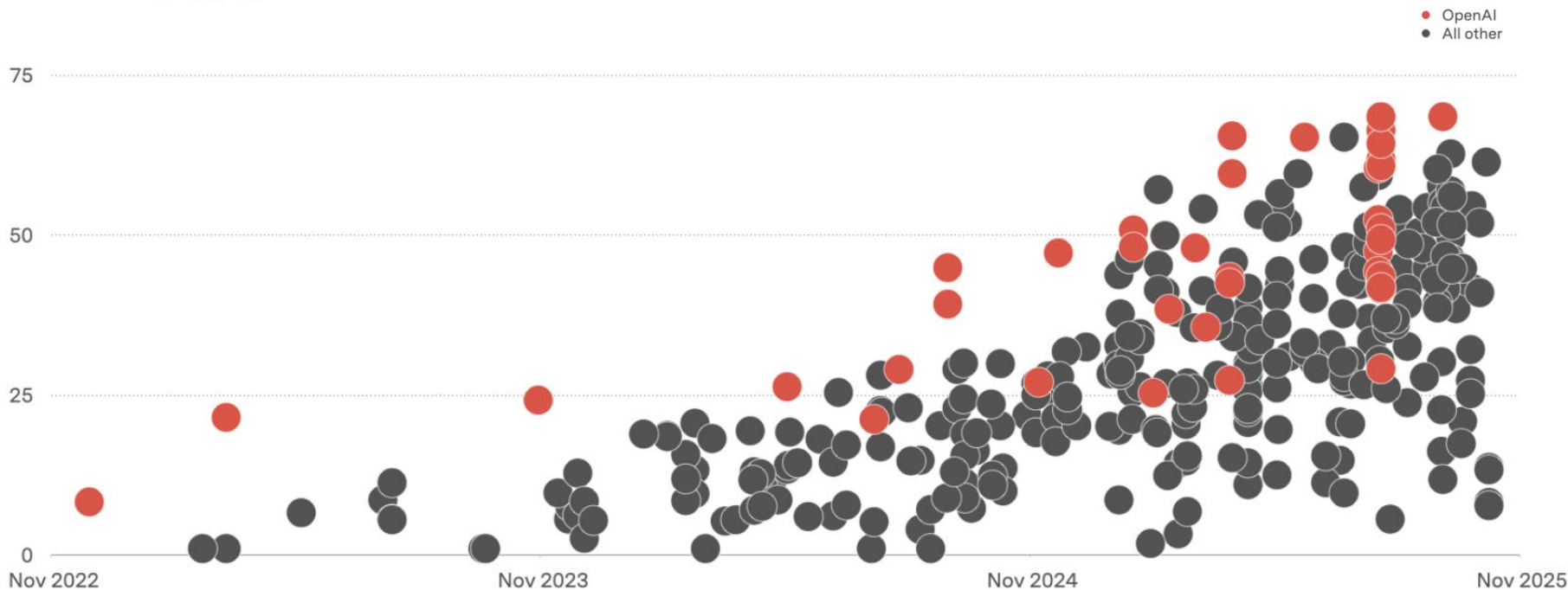
Compare OpenRouter token share by model author



1. minimax	2.07T 23.2%	6. moonshotai	753B 8.4%
2. google	1.33T 14.9%	7. deepseek	584B 6.5%
3. anthropic	1.09T 12.2%	8. x-ai	439B 4.9%
4. openai	899B 10.1%	9. arcee-ai	259B 2.9%
5. z-ai	786B 8.8%	10. others	722B 8.1%

The model (and benchmark) fields are crowded

Models by aggregate benchmark score



Use cases inform relevance of benchmarks

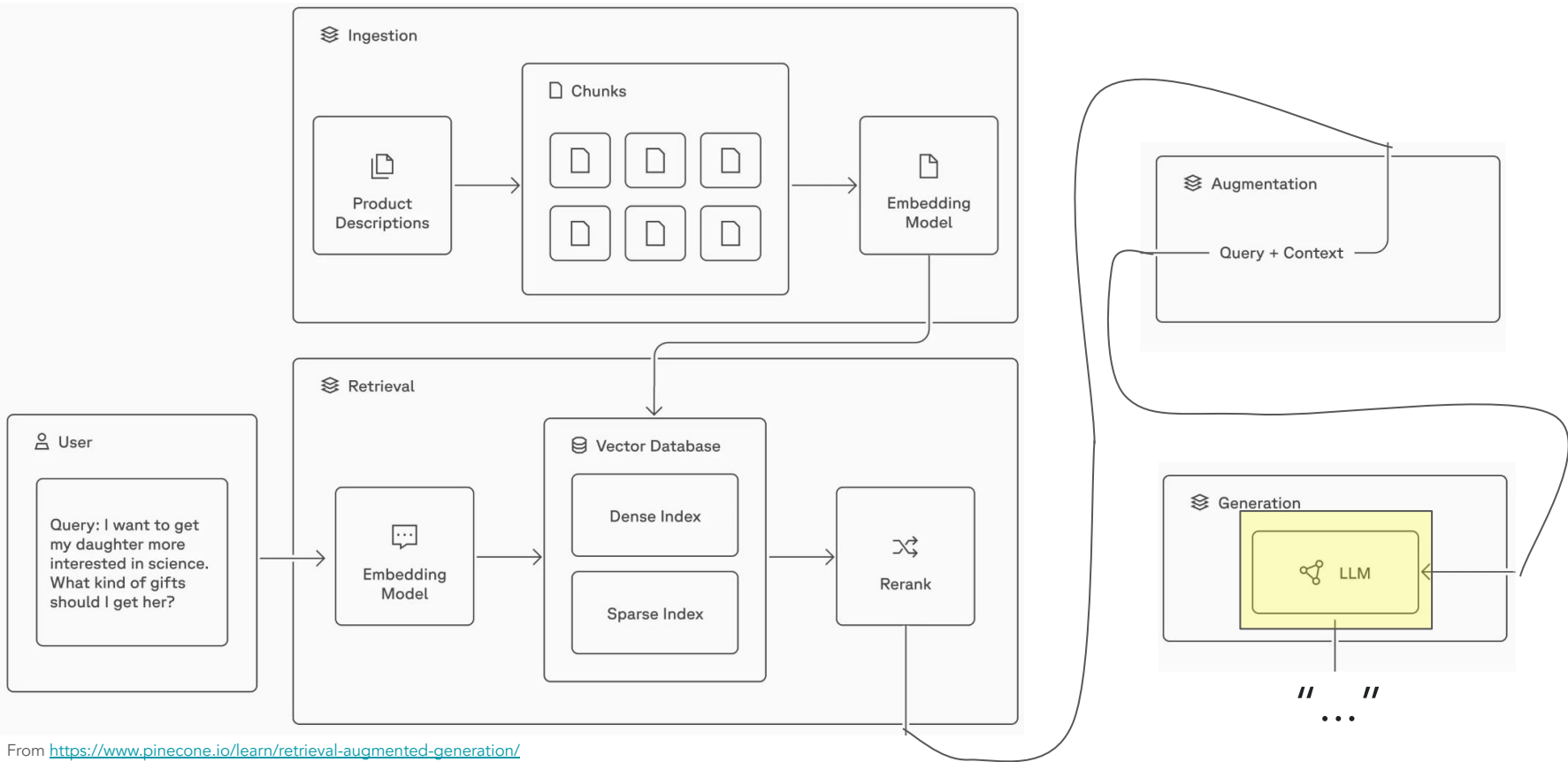


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HREF		✗	0	✓	N/A	Winrate	

From <https://arxiv.org/pdf/2411.15124> , <https://deepmind.google/models/gemini/> , <https://artificialanalysis.ai/>

Model	Percentage of Correct Answers
Claude 4.5 Opus	74%
GLM-4.5	66%
Claude 4.5 Sonnet	62%
Claude 4.5 Haiku	52%
Mistral Sonnet	52%
K2 Think V2	48%
Claude Opus 4.5 (max)	40%
Grok 4	36%
Kimi K2.5	36%
KAT-Coder-Pro V1	34%
GPT-5.2 (high)	29%
Codex (GPT-4)	28%
Claude Opus 4.5	24%
GPT-5.2 (high)	22%
Deepseek	18%
Vesit	17%
Nemotron 3	15%
Mistral Large 3	12%
Llama 4 Maverick	12%
MiniMax-M2.5	12%
Gemini 3 Pro	12%
Pro Preview	12%
K-EAONE	12%
Qwen3.5 327B	12%
Qwen3 72B	12%
A2B 255B	10%
gpt-oss-120B	10%
Nova (high)	10%
Flan 3 (medium)	9%
Flan 3 (high)	9%
gpt-oss-20B (high)	7%

Successful RAG depends on more than LLM



Successful RAG depends on more than LLM

Can the system be effectively deployed and maintained? Is its use encouraged?



Do the users understand it, and how and where to use it?



How are the data represented? How are they being retrieved?



Do the data exist? Are they accessible? Are they clean? Are they correct and up to date?



The winning curling team was from _____

Aligned AI use cases

Aligned AI

Case Study: Airline customer service



AIR CANADA

“On the airline’s website, Moffatt engaged with an AI chatbot, which responded that there is a discount...”

Anyone seeking a reduced fare could allegedly submit their ticket within ninety days of issuance through an online form and receive the lower bereavement rate.”

Case Study: Airline customer service

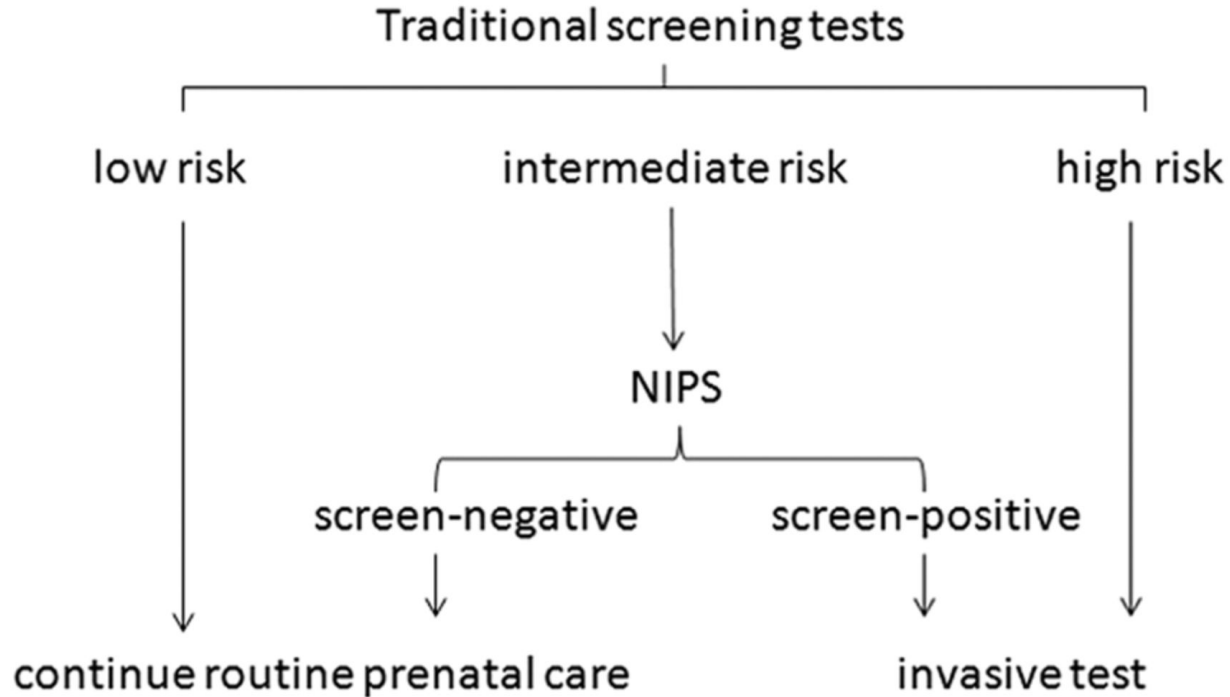
Unfortunately, the AI chatbot's answer was incorrect. The reference to "bereavement fares" was hyperlinked to a separate Air Canada webpage titled "Bereavement travel" that contained additional information regarding Air Canada's bereavement policy. The webpage indicated that the bereavement policy does not apply to requests for bereavement consideration after travel was completed. Accordingly, when Moffatt submitted his application to receive a partial refund of his fare, Air Canada refused. After a series of interactions, Air Canada admitted that the chatbot had provided "misleading words." The representative pointed out the chatbot's link to the bereavement travel webpage and said Air Canada had noted the issue so it could update the chatbot.

Case Study: Airline customer service

Moffatt then sued Air Canada for having relied on its chatbot, which the Tribunal determined was an allegation of negligent misrepresentation. Air Canada alleged that the correct information could have been found elsewhere on its website and argued that it could not be liable for the AI chatbot's responses. ⁶ Strangely, Air Canada endeavored to argue that the chatbot was a separate legal entity that is responsible for its own actions.

The Tribunal ultimately found in favor of Moffatt. While a chatbot has an interactive component, the Tribunal found that the program was just a part of Air Canada's website and Air Canada still bore responsibility for all the information on its website, whether it came from a static page or a chatbot. As a service provider, Air Canada owed Moffatt a duty of care that was breached by the misrepresentation. Air Canada could not separate itself from the AI chatbot, which was integrated in its own website. Negligence existed as Air Canada did not take reasonable care to ensure that its chatbot provided accurate information. It did not matter if the correct information existed elsewhere. A consumer cannot be expected to double-check information it finds on one part of the website with another. ⁷

Case Study: Medical diagnostics



Case Study: Medical diagnostics



Case Study: Medical diagnostics

about a medical system that measured variables and outcomes in various places and times, conducted statistical analyses, and made use of the results in other situations. The problem had to do not just with data analysis per se, but with what database researchers call *provenance*—broadly, where did data arise, what inferences were drawn from the data, and how relevant are those inferences to the present situation? While a trained human might be able to work all of this out on a case-by-case basis, the issue was that of designing a planetary-scale medical system that could do this without the need for such detailed human oversight.

I'm also a computer scientist, and it occurred to me that the principles needed to build planetary-scale inference-and-decision-making systems of this kind, blending computer science with statistics, and considering human utilities, were nowhere to be found in my education. It occurred to me that the development of such principles—which will be needed not only in the medical domain but also in domains such as commerce, transportation, and education—were at least as important as those of building AI systems that can dazzle us with their game-playing or sensorimotor skills.

Case Study: Klarna



Case Study: Klarna

The change marks a significant reversal for a company that was among the first to partner with OpenAI, with Siemiatkowski previously declaring he wanted Klarna to be the tech giant's "favorite guinea pig."

In 2024, Klarna boasted that its AI assistant could handle the work of 700 customer service agents, managing 75% of customer chats in over 35 languages. The company saw this as a path to reduce its workforce from 5,000 to 2,000 employees through natural attrition.

Case Study: Klarna

Customers reported frustrations with the AI's limitations, leading to a decline in service quality.

But the cost-cutting strategy that helped Klarna navigate the end of the fintech boom – which saw its valuation plummet from a peak of \$45.6 billion to as low as \$6.7 billion – has apparently reached its limits. "As cost unfortunately seems to have been a too predominant evaluation factor when organizing this, what you end up having is lower quality," Siemiatkowski admitted. "Really investing in the quality of the human support is the way of the future for us."

Assignment walkthrough

Aligned AI